



MACHINE LEARNING

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GUI Enhancements

The enhanced GUI application includes the following new features:

- Users can now select different loss functions.
- Added user-defined prior probabilities for the GaussianNB model.
- Missing data handling options can now be selected via the GUI.
- Model parameters are dynamically updated, and data is reprocessed based on user selections.
- The model visualization panel has been improved.

Missing Data Handling Methods and Their Effects

This section examines the impact of missing data handling methods on model performance.

4.1 Missing Data Handling Methods

- **Mean Imputation:** Filling missing data with the mean value
- **Interpolation:** Predicting missing data through interpolation between data points
- **Forward Fill / Backward Fill:** Filling missing data based on previous or next values

4.2 Impact of Missing Data Handling Methods on Model Performance

The effect of missing data handling methods on model accuracy was observed as follows:

- **Mean Imputation and Interpolation** helped maintain model accuracy more consistently.
- **Forward/Backward Fill** caused slight deviations in model accuracy in some datasets.
- **User-defined prior probabilities for GaussianNB** significantly influenced model decisions.

Results and Screenshots

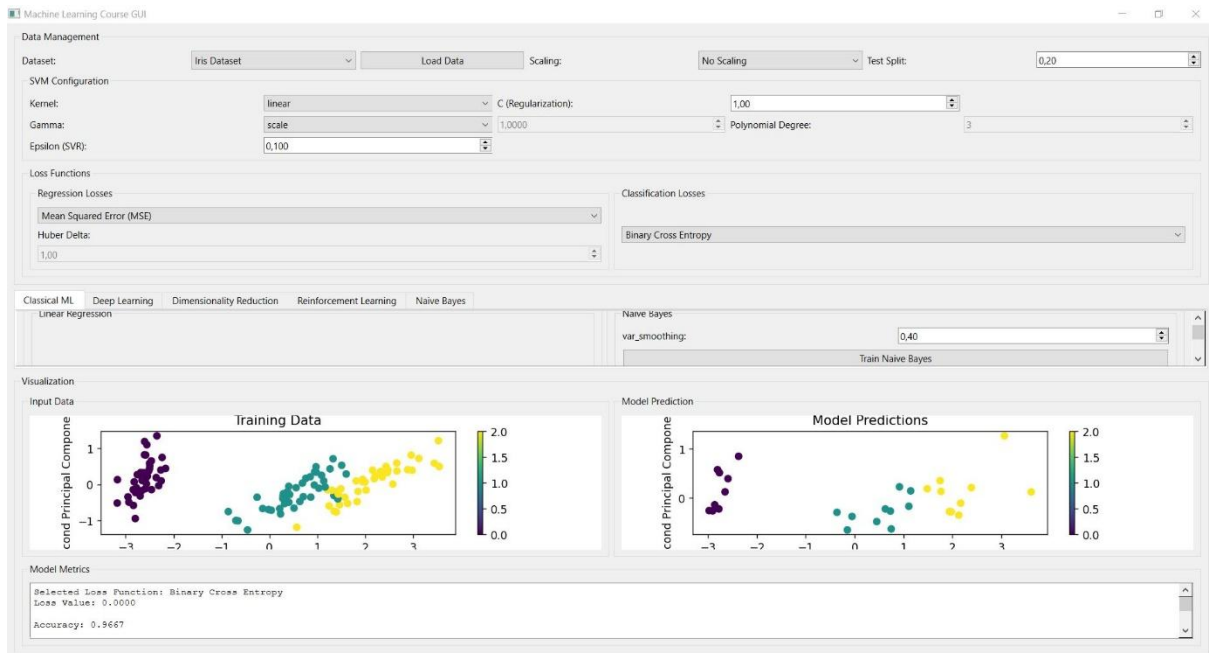


Figure 1

Using the Iris Dataset

The SVM model was trained using the Iris dataset. Model configuration options and loss functions were set, achieving 96.67% accuracy after training. The visualization panel compares training data and model predictions.

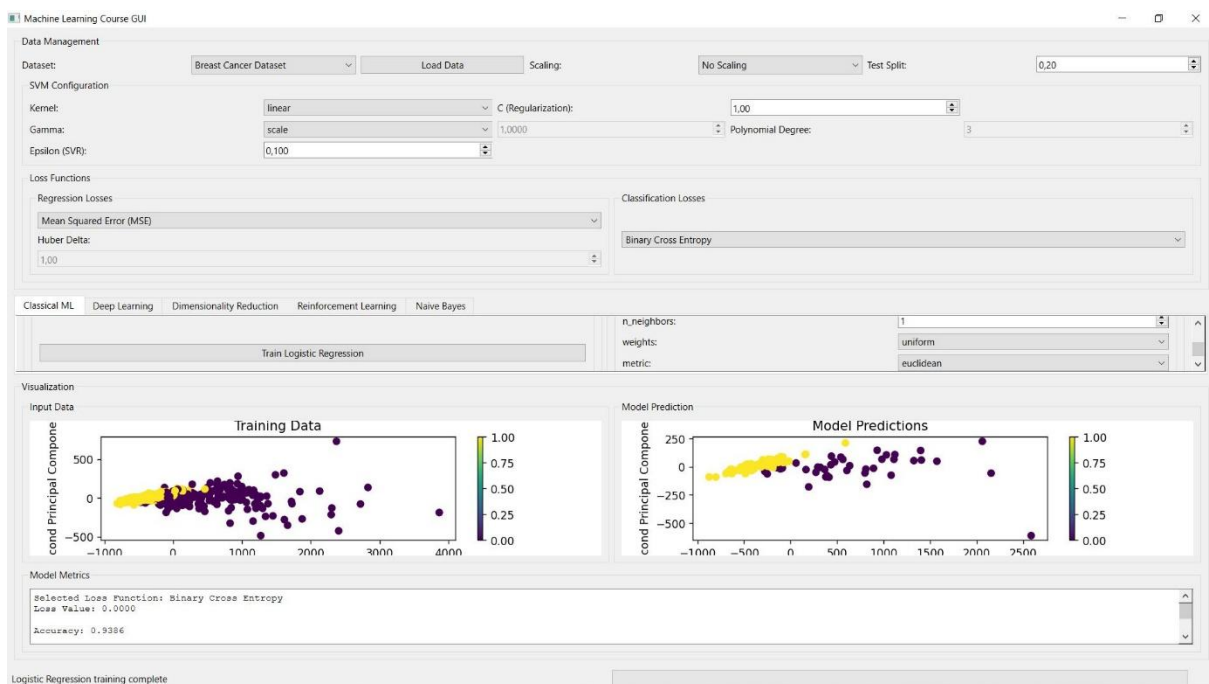


Figure 2

Using the Breast Cancer Dataset

The logistic regression model was trained with the Breast Cancer dataset. Users can customize model and data processing options. The training resulted in 93.86% accuracy. Graphs compare model predictions with training data.

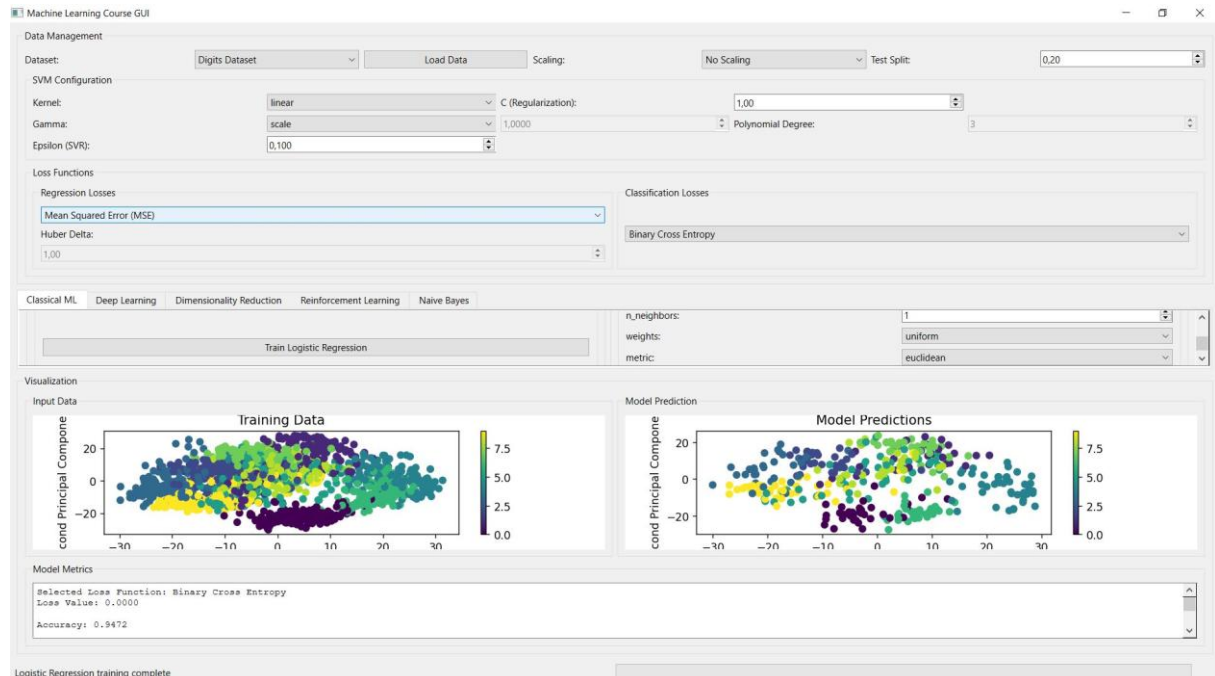


Figure 3

Using the Digits Dataset

Model training was completed with the Digits dataset. Regression and classification options were utilized. The training achieved 94.72% accuracy. The visualization panel displays model predictions and training data.

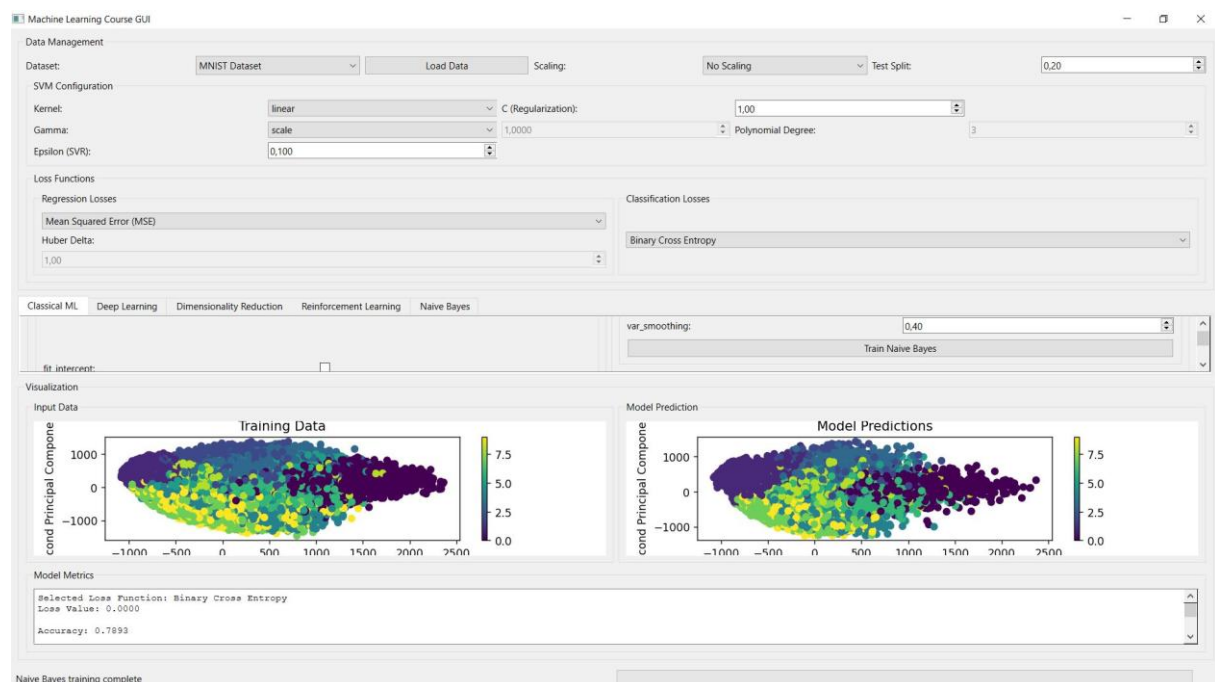


Figure 4